AI for Lunar Olivine Research

Lunar Petrography

The Moon is my beautiful impossibility.

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Tutorial • Colab Notebooks • Lunar Petrographic Exhibit

This guide is built around **what you actually see in lunar thin sections**,and ties each observation to the petrogenetic story it tells.

Contents

1 Optical Petrography

1.1 Petrographic microscope

- lower polarizer
- upper polarizer/analyzer
- rotating stage

1.2 Thin sections

The 30 µm standard exists because at that thickness quartz and feldspar reach a known first-order interference color; thickness errors shift every color you read. Lunar sections are often uncovered and polished so the same chip can be examined in reflected light and by electron microprobe.

1.3 PPL

PPL means plane-polarized light.

1. Relief
2. color
3. Pleochroism
4. Cleavage and fractures
5. Habit
6. Inclusions
7. Alteration

Lunar rocks are essentially unweathered, so alteration is almost always shocked rather than aqueous — a key difference from terrestrial petrography.

1.4 XPL

XPL means cross-polarized light.

1. Isotropy vs. anisotropy
2. Interference color and birefringence order
3. Extinction type (parallel, inclined, symmetrical, undulatory), Extinction angle
4. Twinning

5. Zoning

1.5 Relected light

Reflected light microscopy is essential for the petrographic characterization of opaque and semi-opaque phases in Apollo lunar thin sections, where transmitted light alone is insufficient. In reflected light, ilmenite typically exhibits high reflectivity with a characteristic bluish-white tone, whereas troilite appears slightly lower in reflectance and may show anisotropic etching or tarnish-like effects depending on sample preparation. Native iron, often present as nanophase or larger metallic blebs associated with impact processes and space weathering, is highly reflective and isotropic. Reflected light is particularly powerful for interpreting lunar impact histories, as it reveals melt coatings, shock-produced metal segregation, and regolith maturation features such as agglutinitic glass and submicroscopic iron.

2 Lunar Context

2.1 The Apollo sample collections

Six landing sites returned 382 kg of rock, soil, and core. Each site biases the collection:

1. Apollo 11 and 17 sampled high-Ti mare basalts
2. Apollo 12 and 15 low-Ti mare basalts and KREEP
3. Apollo 14 the Fra Mauro (Imbrium ejecta) breccias
4. Apollo 16 the feldspathic highlands
5. Apollo 15 and 17 also reached highland massifs and pyroclastic glass deposits.

Samples are curated and subdivided at JSC; every chip and section carries a generation number tracing it to a parent.

2.2 The LMO framework

1. Early global melting $\rightarrow$ crystallization $\rightarrow$ dense mafic cumulates sink, buoyant plagioclase floats to form the ferroan anorthosite (FAN) primary crust.
2. Residual liquid concentrates incompatible elements into KREEP (K, REE, P).
3. Later Mg-suite magmas (norites, troctolites, dunites) intrude the crust.
4. Much later, partial melting of mantle cumulates produces mare basalts that flood the basins.

This single model organizes nearly every thin section you will see.

3 Lunar Mineralogy under the Microscope

Plagioclase is the most abundant highland mineral; major in basalts. Lunar plagioclase is low relief, low (first-order) birefringence, polysynthetic twinning.

Pyroxenes are dominant mafic in mare basalts. Augite: high relief, second-order interference colors, inclined extinction, pale brown in PPL. Orthopyroxene: parallel extinction, lower birefringence; common in Mg-suite norites.

Olivine: High relief, no cleavage (irregular/conchoidal fracture), bright second-to-third-order interference colors, parallel extinction, bright in PPL. Phenocrysts in some mare basalts (Apollo 12, 15) and the essential mafic phase of troctolites and dunites.

Oxides need to be examined in reflected light. Ilmenite: opaque in transmitted light; abundant in high-Ti basalts (Apollo 11, 17). Reflected light: grayish, distinctly anisotropic; lath or skeletal habit.

Glass: Interstitial mesostasis glass, impact glass, and primary pyroclastic glass beads. All isotropic (dark in XPL); color in PPL ranges from clear to green, orange, brown.

4 Igneous Textures & Mare Basalts

4.1 Texture vocabulary

Ophitic / subophitic: pyroxene oikocrysts enclosing plagioclase laths, the signature texture of many slowly cooled mare basalts.

Intergranular / intersertal: mafic grains (or glass) filling between plagioclase laths.

Porphyritic: phenocrysts (olivine or pyroxene) in a finer groundmass — records two-stage cooling.

Variolitic / radiate: fans of plagioclase + pyroxene from rapid (quench) growth.

Vitrophyric: phenocrysts set in glass — very fast cooling, e.g., flow tops or impact melts.

Equigranular / granular: uniform grain size, common in coarser interiors.

4.2 Crystallization sequence

In mare basalts, early oxides (chromite, armalcolite/ilmenite in high-Ti melts) and olivine give way to pyroxene and plagioclase, with silica, phosphates, K-glass, metal, and troilite in the last-crystallizing mesostasis. Reading the order from inclusion relationships and groundmass position is a core skill.

5 Resources

1. Lunar Sourcebook
2. Lunar Sample Compendium
3. Lunar Sample Atlas / Apollo Sample Catalogs
4. Virtual Microscope
5. NASA Lunar Sample / Thin Section education loan program